Natural Language Processing

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CSCI 5832

Boulder NLP

Today

- More morphology

- Vocabularies
- Corpora
- Text normalization/segmentation
- Subword tokenization

Regular Inflection in English

- Nouns are simple
  - Markers for plural and possessive
    - -s
- Verbs are only slightly more complex
  - Markers appropriate to the tense of the verb
    - -ed, -ing, -s
- And that’s pretty much it
  - Other languages can be quite a bit more complex… more complex verb inflection, inflecting nouns, adjectives, etc.
  - A consequence of this is that approaches that work well enough English will <u>not</u> work for many of the world’s other languages.

Derivational Morphology

Derivational morphology is the messy stuff that no one ever taught you (at least not very well). It is characterized by

- Wide variety of affixes
  - -al, -tion, -ee, -er, -ant, -ness…
  - anti-, un-, non-…
- Quasi-systematic rules
- Irregular meaning change
- Changes of word class

Derivational Example: C*ompute*

Consider the verb *compute*

- Compute -> computer
- Computer -> computerize
- Computerize -> computerization

Derivational Examples

-ation	computerize	computerization
-ee	appoint	appointee
-ant	contest	contestant
-er	serve	server
-ant	serve	servant
-ness	fuzzy	fuzziness

Verbs and Adjectives $\rightarrow$ Nouns

Derivational Examples

-al	computation	computational
-able	embrace	embraceable
-ive	impress	impressive
-less	clue	clueless

Nouns and Verbs $\rightarrow$ Adjectives

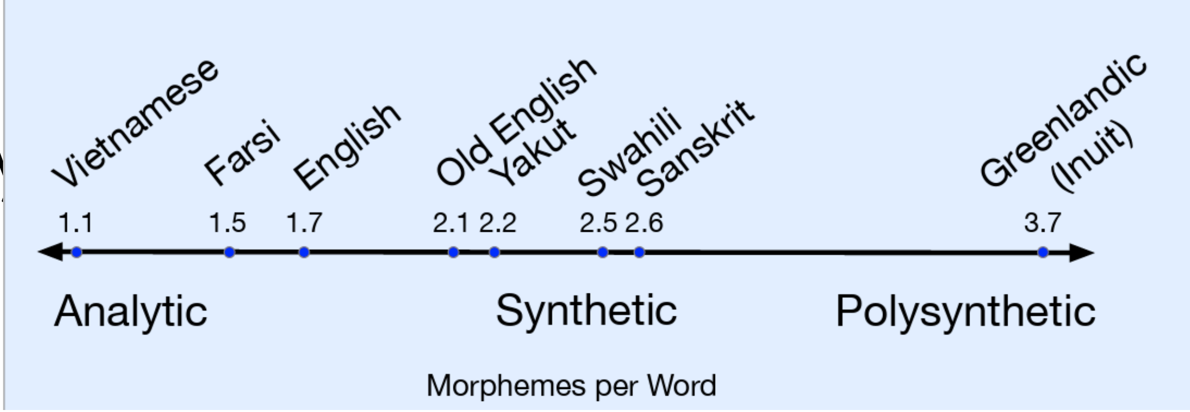
Language Classification

There are several ways to “classify” languages: divide them up into categories based on some characteristics.

- Genetically: Group languages based on their family history (romance, indo-European etc.
- Areal: Group languages based on where they occur (or did occur) geographically.
- Typological: Group languages based on aspects of their grammar (morphology, phonology, syntax, etc).

Such groupings can be useful in NLP when trying to deal with new languages with little training data.

Morphological Ty



Morphological typology g
morphology largely along

How many morphemes per word?

- Isolating vs. Synthetic
  - Isolating languages tends toward one morpheme per word
  - Synthetic languages can jam lots of morphemes into a single word
    - Polysynthetic languages jam entire sentences into single words

How hard is it to find the morphemes in a word?

- Agglutinating vs. Fusional
  - Agglutinating languages form words by concatenating stems with affixes to form words (with prefixes, infixes or suffixes). Fusional languages allow the morphemes to change their surface form upon inflection, making it difficult to isolate them.

Morphological Typology

English tends to be on the isolating end of the spectrum. But exhibits both fusional and agglutinating properties.

Walk -> walked, walking, walks

but

Catch -> Caught

Right -> Rang

Cut -> Cut

Agglutinating Example

Turkish (synthetic/agglutinating)

- Uygarlastiramadiklarimizdanmissinizcasina
- *`(behaving) as if you are among those whom we could not civilize'*
- Uygar `civilized' + las `become'

  + tir `cause' + ama `not able'

  + dik `past' + lar 'plural'

  + imiz 'p1pl' + dan 'abl'

  + mis 'past' + siniz '2pl' + casina 'as if'

Complex Morphology

While Arapaho is synthetic and agglutinating language

- Ne'toukutooxebei3i'

Then they tied up their horses.

In this example, all the words appear as part of the verb – including both arguments of the verb *tie* (they, and horses) as well as the tense. Further the form of '<u>horse</u>' in this utterance is distinct from the standalone form for the root horse.

Computational Morphology

Morphological analysis takes a surface form and returns the stem with a set of morphological features rather than just segmenting the surface form.

- Cats → Cat+PL   Ate → eat+PAST
- Generation goes in the other direction.

Computationally capturing the morphology of any language is very very difficult and involves the use of either probabilistic/weighted state machines or neural networks.

But for many languages, like English and Chinese, you can get away with ~~stupid~~ simple approaches.

- This has implications for how well languages that are not like English/Chinese get treated in commercial applications.

Vocabulary

A vocabulary is the set of words that a system has basic access to. The minimal units of processing in an input that has some semblance of meaning attached to it.

What should such a vocabulary (or lexicon) consist of?

- All the words in a language?
  - All the words in some comprehensive dictionary?
- Or some subset?
  - The words needed for a particular application?
    - How many?
    - How do we determine the right list?
    - Is it fixed or can we add new words?

From Corpora to Vocabularies

Most NLP approaches assume a corpus-based approach to developing a vocabulary.

1. Collect a big (representative) bunch of text.
2. Figure out the words that appear in it.
3. That’s your vocabulary.

Corpora and Text Normalization

Every NLP application needs to do some form of *text normalization*. Not terribly glamorous, but critical to further processing and often the source of numerous headaches.

- Segmenting running text into sentences
  - Or dialogs into “turns”
- Tokenizing sentences into words
- Normalizing word formats
- Recognizing different scripts/languages

- Producing a consistent stream of tokens given raw text inputs.
- Determining a fixed useful vocabulary
- Dealing with unknown words

Simple Tokenization in UNIX

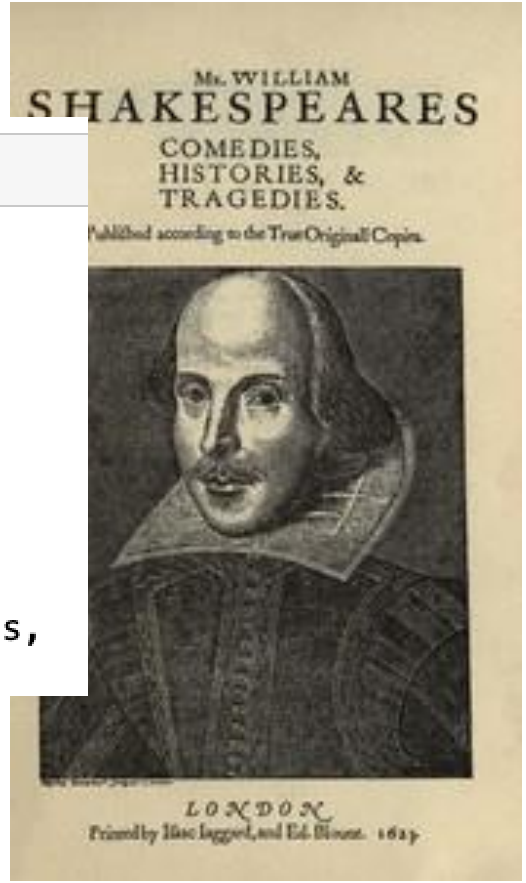
```
cat shakes.txt | head
```

```
THE SONNETS

by William Shakespeare


                          1
  From fairest creatures we desire increase,
  That thereby beauty's rose might never die,
  But as the riper should by time decease,
  His tender heir might bear his memory:
  But thou contracted to thine own bright eyes,
```

- Given a corpu output all the their associat
  - Nothing fa
- Turns out tha commands to processing op pretty easy task to do in a crude fashion.



Inspired by Ken Church's UNIX for Poets.

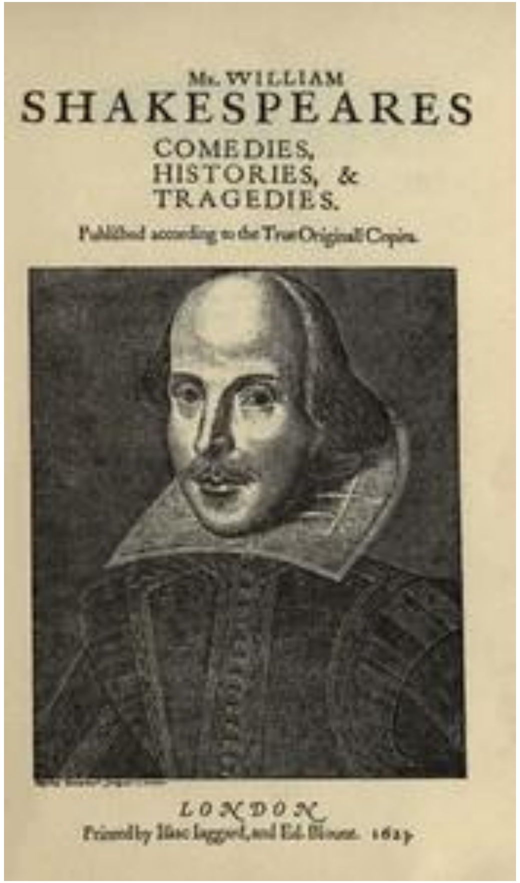
```
wc shakes.txt
```

```
  122394  883319 5450367 shakes.txt
```

Lines Words Bytes

```
tr -sc 'A-Za-z' '\n' < shakes.txt | head
```

```
THE
SONNETS
by
William
Shakespeare
From
fairest
creatures
we
desire
```



```
tr -sc 'A-Za-z' '\n' < shakes.txt | sort | head
```

```
A
A
A
A
A
A
A
A
A
A
```

```
tr -sc 'A-Za-z' '\n' < shakes.txt | sort | uniq | head
```

```
A
AARON
ABBESS
ABBOT
ABERGAVENNY
ABHORSON
```

```
tr -sc 'A-Za-z' '\n' < shakes.txt | sort | uniq | wc -l
```

```
29152
```

```
tr 'A-Z' 'a-z' < shakes.txt > shakesdown.txt
```

```
tr -cs "a-z" '\n' < shakesdown.txt | sort | uniq | wc -l
```

```
23512
```

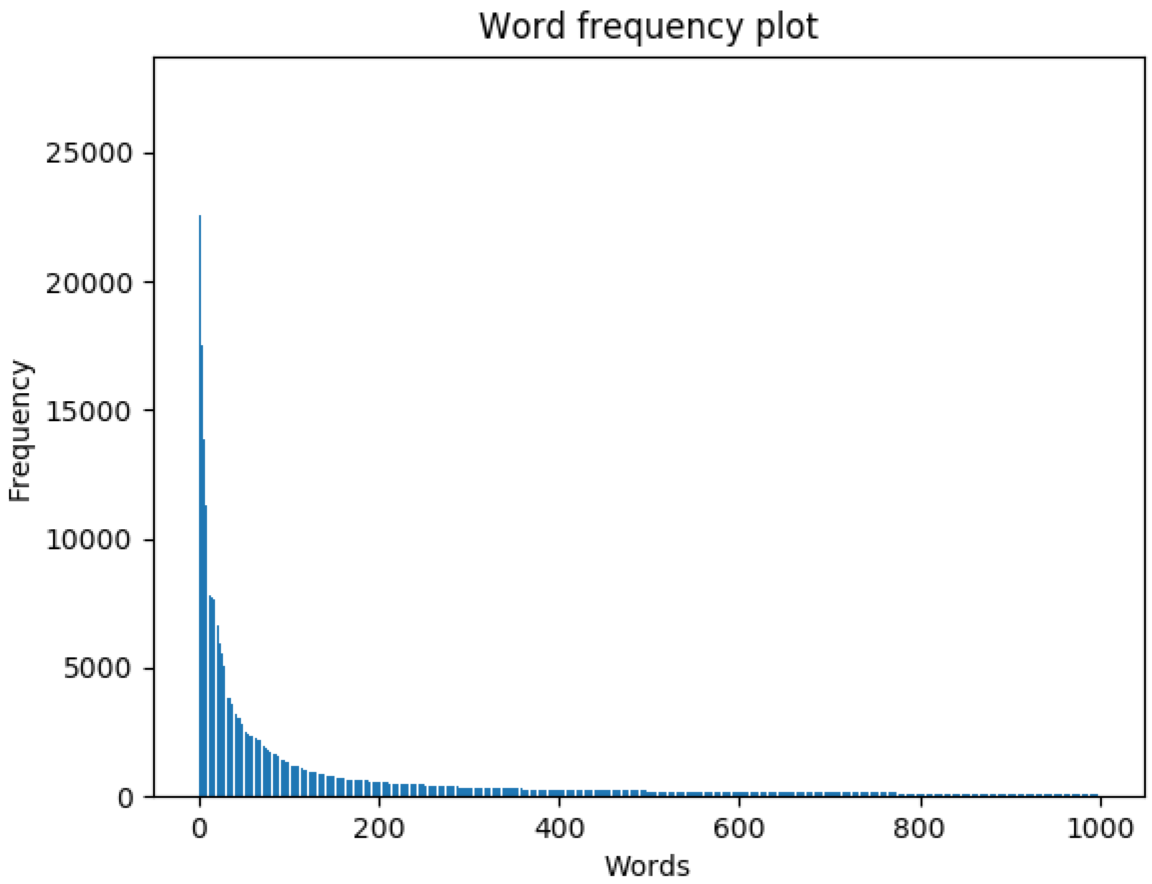
```
tr -sc 'a-z' '\n' < shakesdown.txt | sort | uniq -c | head
```

```
14725 a
   97 aaron
    1 abaissiez
   10 abandon
    2 abandoned
    2 abase
    1 abash
   14 abate
    3 abated
    3 abatement
```

```
tr -sc 'a-z' '\n' < shakesdown.txt |  sort | uniq -c | sort -rn |  head -25
```

```
27378 the
26084 and
22538 i
19771 to
17481 of
14725 a
13826 you
12490 my
11318 that
11112 in
9319 is
8960 d
8512 not
7791 with
7778 me
7725 it
7721 s
7655 for
6897 be
6859 his
6679 he
6657 your
6608 this
6277 but
5902 have
```

What's up with these?



Word frequency plot
Frequency
Words
25000
20000
15000
10000
5000
0
0
200
400
600
800
1000

Corpora

Words don’t just appear out of nowhere.

- In building NLP systems we make use of representative texts (Corpora) – either generically, for use with diverse applications, or designed for a specific domain or application.
- <u>But</u> texts are produced by specific writer(s), at a specific times, in a variety of a languages, for specific functions. All of which combine combinatorically. All of these factors matter for system design.

Dimensions of Corpora

- Language
  - 7097 languages in the world (not all have written forms)
  - Dialects, creoles, pidgins, etc.
- Provenance
  - How was it obtained? Was there consent? Is it legal to possess/distribute?
- Genre
  - Newswire, fiction, non-fiction, scientific articles, Wikipedia
- Demographics
  - Writers’ age, gender, race, socioeconomic status, etc.
- Medium
  - Spoken, written, captioned, video
- Writing Systems
  - Over 350 systems in active use

How Many Words Are There?

N = number of mentions

V = vocabulary = set of types

$|V|$ is the size of the vocabulary

How Many Words Are There in English?

N = number of tokens

V = vocabulary = set of types

$|V|$ is the size of the vocabulary

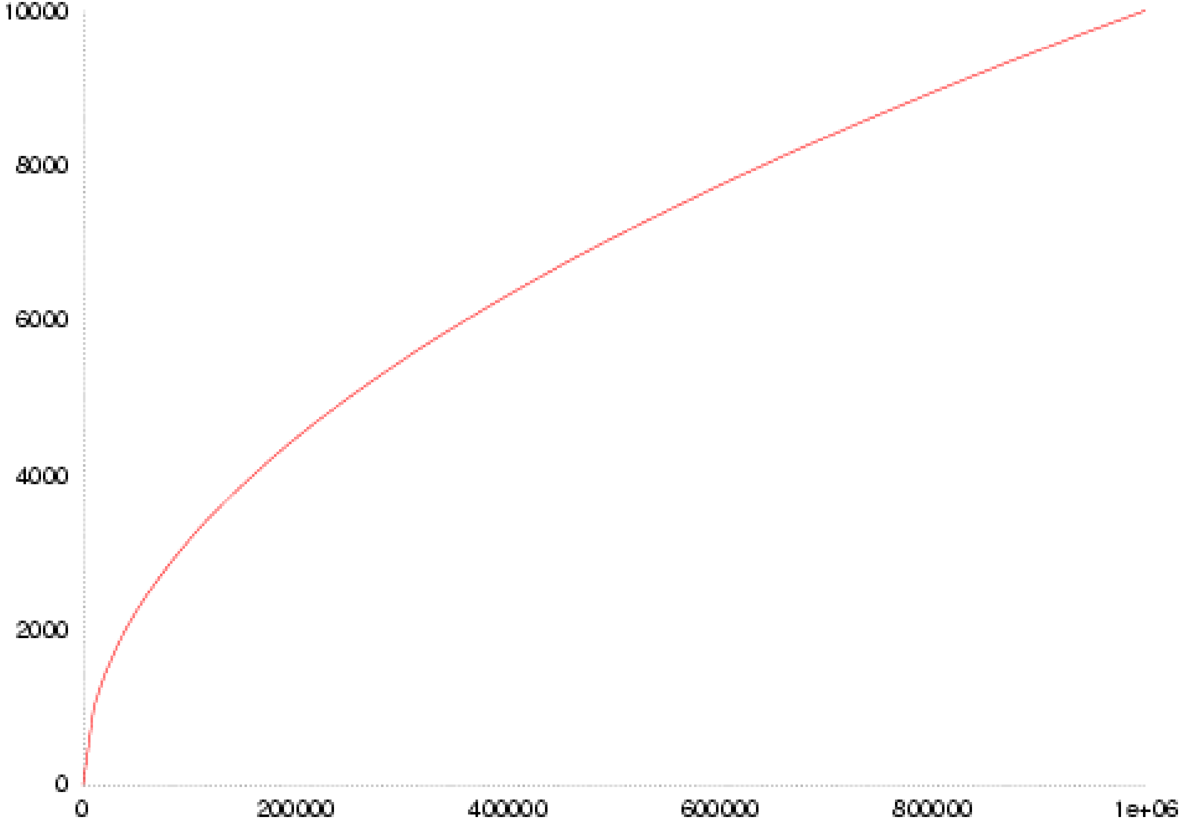
	Tokens = N	Types = |V|
Switchboard phone conversations	2.4 million	20K
Shakespeare	884,000	23K
Common Crawl (2017)	600B	1.9M
Google N-grams Corpus	1 trillion	13M

Heaps' Law $|V| = kN^{\beta}$

Where *k* and $\boldsymbol{\beta}$ are free variables. Usually, k is between 10 and 100 and $\boldsymbol{\beta}$ is between 0.6 and 0.7.

- Roughly, this means that the size of the vocabulary grows <u>somewhat faster</u> than the square root of the corpus size
- Which really means that the rate of growth of the vocabulary slows as a corpus grows in size but <u>never flattens out</u>.

Heaps' Law



$$|V| = kN^{\beta}$$

Issues in Normalization/Tokenization

- Finland’s capital → Finland Finlands Finland ’s
- what’re, I’m, isn’t → What are, I am, is not
- Hewlett-Packard → Hewlett Packard
- state-of-the-art → state of the art
- Lowercase → lower-case lowercase
- Hong Kong → One word or 2?
- m.p.g. → keep or discard?
- IRA , I.R.A.? → Expand out? How?

Sentence Segmentation

Tokenization, morphology and further syntactic processing first requires sentence segmentation

- In English, punctuation is used to mark sentence boundaries
  - !, ? are relatively unambiguous
  - Period "." is quite ambiguous
    - Abbreviations like Inc. or Dr.
    - Acronyms like U.N.
    - Numbers like .02% or 4.3

Sentence Segmentation

Stocks were up as advancing issues outpaced declining issues on the NYSE by 1.5 to 1. Large- and small-cap stocks were both strong, while the S.&P. 500 index gained 0.46% to finish at 2,457.59. Among individual stocks, the two top percentage gainers in the S.&P. 500 were Incyte Corporation and Gilead Sciences Inc.

Multilingual Issues in Normalization

Thus far our discussions have focused on written English texts – typically newswire as a genre. But English isn’t the only language out there and even applications focused on English need to deal with other languages. Among the issues to be faced:

- Code switching
- Writing systems
- Complex morphology
- Word segmentation
- Sentence segmentation

Code Switching

Mixing different languages in the same utterance

Spanish + English:

Por primera vez veo a @username actually being hateful! It was beautiful:)

[For the first time I get to see @username actually being hateful! it was beautiful:)]

Hindi + English:

dost tha or rahega ... dont wory ... but dherya rakhe

["he was and will remain a friend ... don't worry ... but have faith"]

Writing Systems (Scripts)

Latin

Chinese

Arabic

Devanagari

Bengali

Arabic

Cyrillic

Greek

Hebrew

Etc.

Segmentation Issues

Chinese/Thai/Japanese do not reliably use spaces between words

- **莎拉波娃**现在居住在美国东南部的佛罗里达。
- **莎拉波娃** 现在 **居住 在** **美国** 东南部 **的** **佛**罗里达
- *Sharapova now lives in US southeastern Florida*

HW 1 Part 1: 50 Points

- How many words do you know in your native language?
  - Due Tuesday 8/1 by 11:59PM
  - Submit via Canvas
  - Your answer and a writeup explaining your answer.
    - No longer than necessary. 2-3 pages should suffice.
    - Long enough to say something interesting
    - PDF; follow the naming convention specified on the Canvas assignment page

HW 1 Part 1

How many words do you know in your native language?

- In your answer, clearly address the following aspects of this question.
  - “how many” – how did you interpret the counting task?
  - “words” – what is your operational definition of what a word is?
  - “know” – what constitutes evidence that you know a word?

Next Time

1. Byte-Pair Encoding
2. Language Models